

20/12/24
Friday

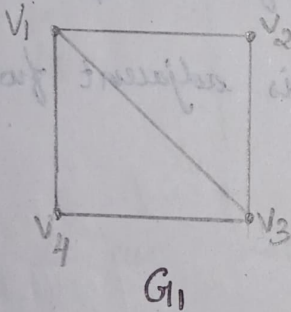
Module-1

Introduction to graphs

Graph:

→ A graph is a pair of (V, E) where V is the finite non-empty set of vertices or nodes & E is a set of edges or arcs, where E is a subset of $V \times V$. V denote G by (V, E) also the edge $e \in E$ is associated with a pair of vertices (U, V) called end vertices.

Eg:



G_1

simple graph

$$V = \{V_1, V_2, V_3, V_4\}$$

$$E = \{(V_1, V_2), (V_2, V_3), (V_4, V_3), (V_4, V_1), (V_1, V_3)\}$$

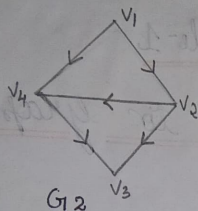
Diagraph [Directed graph]:

→ A diagraph or directed graph G consists of a set of vertices & a set of edges such that each edge 'e' is associated with an ordered pair of vertices i.e. each edge of G has a direction.

NOTE:

In diagraph the order of vertices is important.

Eg:



$$V = \{v_1, v_2, v_3, v_4\}$$

$$E = \{(v_1, v_2), (v_1, v_3), (v_1, v_4), (v_2, v_3), (v_2, v_4), (v_3, v_4)\}$$

NOTE:

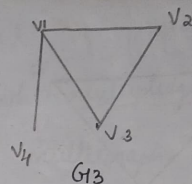
- Let $E = (u, v)$ is an edge in a directed graph then
- u is the initial vertex of e .
 - v is the terminal vertex of e .
 - e is said to be incident from u & incident to v .
 - u is adjacent to v & v is adjacent from u .

Undirected graph:

- An undirected graph G consists of a set of V of vertices & set E of edges & such that each edge is associated with an unordered pair of vertices. In this graph, the direction is not specified.
- Eg: Graph G_1 is an undirected graph.

Degree of a vertex:

The no. of edges connected with a vertex v is called the degree of the vertex v . It is denoted by $d(v)$.



$$d(v_1) = 3$$

$$d(v_2) = 0$$

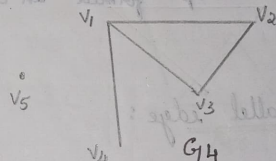
$$d(v_3) = 2$$

$$d(v_4) = 1$$

Isolated vertex:

- A vertex v which has no incident edges is called an isolated vertex.
- Degree of the isolated vertex is zero.

Eg:



v_5 is an isolated vertex.

Null Graph or Empty graph or Trivial graph:

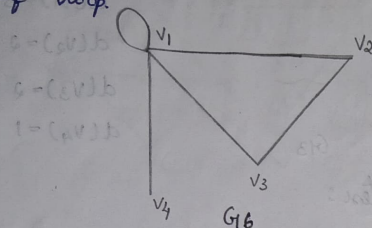
A graph with no edges is called as a null graph or all the vertices in an null graph are isolated vertices.

Loop:

- An edge $e = (v_i, v_i)$ with same end points is called a loop that is an edge of an graph joins, vertex to itself is called a loop.

or self loop.

Eg:



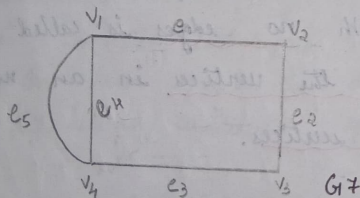
V_1 is the loop. degree of vertex $d(V_1) = 5$.

clearly in G_6 loop is formed at the vertex V_1 & degree of the loop formed in a vertex is taken as 2.

Multiedge or parallel edge:

→ The edges which are joined by same pair of vertices are called parallel edges or multiple edges.

Eg:



Graph G_7 has parallel edges e_4 & e_5 .

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Monday

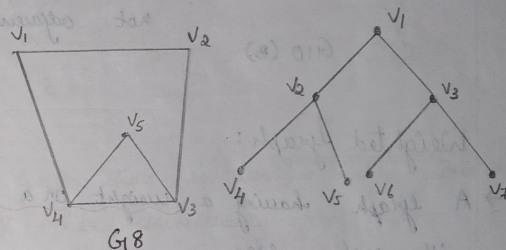
Multigraph:

→ A graph with multiple edges & no loops.

Eg: G_7 is a multigraph.

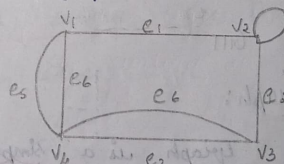
Simple graph:

→ A graph with no parallel edges & loops is called a simple graph.



Pseudo graph:

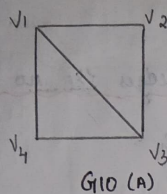
→ A graph which contains parallel edges & loops.
ie pseudograph (implies) → multiple edges + loop.



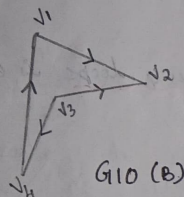
Adjacent vertices:

→ A vertex V_i is adjacent to another vertex V_j if there is an edge from V_i to V_j .

Eg:



clearly (v_1, v_1) (v_2, v_3) are adjacent vertices but v_2 is not adjacent to v_4 .

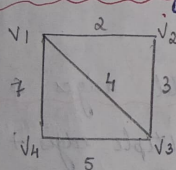


v_1 is adjacent to v_2 but not adjacent to v_4 .

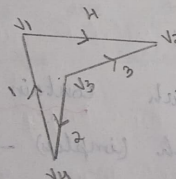
Weighted graph:

→ A graph having a weight or a number associated with each edge

Eg:



G11

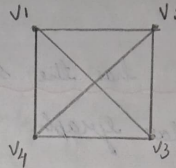


Complete graph:

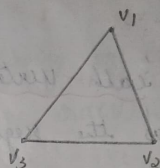
→ A complete graph is a simple graph in which every pair of distinct vertices is connected by a unique edge.

→ A complete graph with n vertices is denoted by ' K_n '.

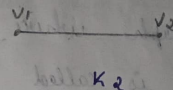
Eg:



K_4



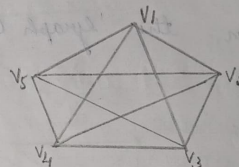
K_3



K_2



K_1



K_5

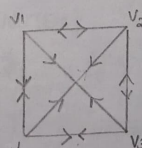
NOTE:

→ In complete graph there is unique edge from any vertex to all other vertices.

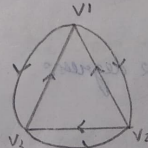
Complete digraph:

→ A complete digraph is a directed graph in which every pair of distinct vertices is connected by a pair of unique edges. [one in each direction] or bi-direction edges.

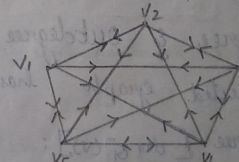
Eg:



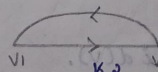
K_4



K_3



K_5



K_2

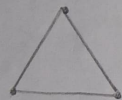


K_1

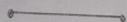
Regular Graph:

- A graph where each vertex has the same degree is called as the regular graph.
- The degree of V , $d(V) = k$ for every vertex of the graph G , then the graph G is called a k -regular graph.

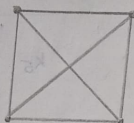
Eg:



2-regular graph



1-regular graph



3-regular graph

NOTE:

- K_n is a regular graph with n vertices then each of the vertices have the degree $n-1$.
- If G is a k -regular graph with n -vertices, then G has $k \times n/2$ edges.

Indegree & Outdegree:

A directed graph has 2 degrees:

→ Indegree [$\deg G^-(v)$]:

No. of edges incident into v .

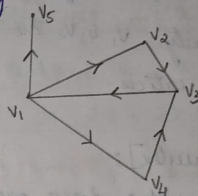
OR

It can be simply denoted as $i.d(v)$.

→ Outdegree:

No. of edges outgoing or leaving from v . It can be simply denoted as, $O.d(v)$.

Eg:



Indegree

outdegree

$$i(v_1) = 1$$

$$O(v_1) = 3$$

$$i(v_2) = 1$$

$$O(v_2) = 1$$

$$i(v_3) = 2$$

$$O(v_3) = 1$$

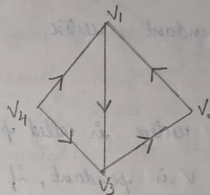
$$i(v_4) = 1$$

$$O(v_4) = 1$$

$$i(v_5) = 1$$

$$O(v_5) = 1$$

Eg:



Indegree

outdegree

$$i(v_1) = 2$$

$$O(v_1) = 1$$

$$i(v_2) = 1$$

$$O(v_2) = 2$$

$$i(v_3) = 2$$

$$O(v_3) = 0$$

$$i(v_4) = 0$$

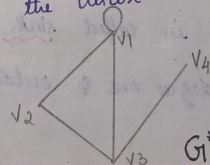
$$O(v_4) = 2$$

$$\text{Sum of indegree} = i.d(v_1) + i.d(v_2) + i.d(v_3) + i.d(v_4) \\ = 2 + 1 + 2 + 0 = 5$$

$$\text{Sum of outdegree} = O.d(v_1) + O.d(v_2) + O.d(v_3) + O.d(v_4) \\ = 1 + 1 + 1 + 2 = 5$$

NOTE:

In a graph the loop at a vertex has 2 incident edges for the vertex v .



$$d(v_1) = 4$$

$$d(v_4) = 1$$

$$d(v_2) = 2$$

$$d(v_3) = 3$$

G^*

Odd & even vertices:

A vertex of a graph G is called odd, if its degree is odd. And it is called even if its degree is even.

Eg: From the above graph clearly v_1 & v_2 are even vertices & v_3 & v_4 are odd vertices.

Pendant vertex [leaf or end vertex]:

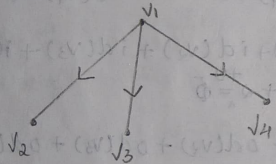
★ A pendant vertex is a vertex with a degree one.

Eg: In graph G^* , v_4 is a pendant vertex.

NOTE:

→ Edge incident to a pendant vertex is called pendant edge.

→ In directed graphs, a vertex v is pendant, if indegree of v + outdegree of $v = 1$. $\text{id}(v) + \text{od}(v) = 1$



Clearly v_2, v_3, v_4 are pendant vertices.

→ The sum of $\text{id}(v)$ and $\text{od}(v)$ of the vertex is called the total vertex degree of the vertex.

→ A vertex with zero indegree is called source.

→ A vertex with zero outdegree is called sink.

→ A loop contributes an indegree one & outdegree one in a digraph.

→

$$\sum \deg G^-(v) = \sum \deg G^+(v) = \text{no. of edges.}$$

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Friday

Handshaking theorem [First theorem of G, T]:

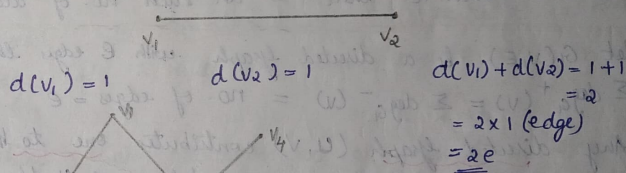
Let $G = (V, E)$ be an undirected graph then $\sum d(v) =$

$$2|E| \quad \text{or} \quad 2e = 2 \times \text{no. of edges.}$$

★ Proof: sum of degree of all vertices equals to twice the no. of edges.

Let $[v_1, v_2]$ be an edge of G & it contributes a count of one of each degree of v_1 & degree of v_2 & consequently a count twice to the sum of the degrees. i.e. when the degree of the vertices are summed each edge is counted twice. Therefore $\sum d(v) = 2 \times |E| = 2 \times \text{no. of edges}$.

Eg:



Corollary:

In any graph $G = (V, E)$ there is an even no. of odd vertices.

OR

In any graph $G(V, E)$, the no. of vertices of odd degree must be even.

Proof:

Let U be the set of odd vertices & V be the set of even vertices, then by the 1st theorem of graph theory $\sum d(v) = 2e$

$$\text{ie } \sum d(u) + \sum d(v) = 2e \quad [\text{because it's 2 times a integer}]$$

Since V is even, $d(v)$ is even. Therefore $\sum d(v) = \text{even}$

Therefore we have $\sum d(u) + \text{even} = \text{even}$

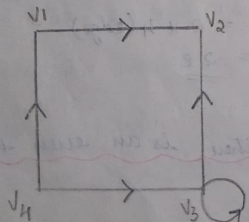
$$\Rightarrow \sum d(u) = \text{even} \quad [\text{log total sum is even}]$$

But U is odd, therefore $d(u)$ is odd. But we have $\sum d(u) = \text{even}$ ie there is an even no. of odd vertices

Let $G(V, E)$ be a directed graph with E edges then $\sum \deg_G^+(v) = \sum \deg_G^-(v) = \text{no. of edges} = e$

Any directed graph (U, V) contributes one to the indegree of V & one to the outdegree of U . Also loop at V contributes one indegree & one outdegree. Then clearly from above graph we have the result.

Eg:



In-deg	out-deg
$id(v_1) = 1$	$od(v_1) = 1$
$id(v_2) = 2$	$od(v_2) = 0$
$id(v_3) = 2$	$od(v_3) = 2$
$id(v_4) = 0$	$od(v_4) = 1$

Therefore we have $\sum \text{indegree} = \sum \text{outdegree} = \text{no. of edges}$

1. Show that degree of vertex of a simple graph G with n vertices cannot exceed $n-1$.

Ans: Let G be a simple graph with n vertices $v_1, v_2, v_3, \dots, v_n$, with e edges, then by first theorem of G-T, we have $\sum d(v) = 2e$ or $d(v_1) + d(v_2) + \dots + d(v_n) = 2e$. Since G is a simple graph with n vertices each vertex in G

Given that G is a simple graph, therefore it has no loops & parallel edges. v be a vertex in G . Since there is no loops & parallel edges v can be adjacent to almost all the remaining $n-1$ vertices of G . Hence the degree of v must be at most $n-1$ ie $d(v) \leq n-1$.

2. Show that maximum no. of edges in a simple graph with n vertices is $\frac{n(n-1)}{2}$.

Ans: Let G be a simple graph with n vertices $v_1, v_2, v_3, \dots, v_n$ with e edges. Then by first theorem of G-T we have $\sum d(v) = 2e$ or $d(v_1) + d(v_2) + \dots + d(v_n) = 2e$. ie $2e \leq (n-1) + (n-1) + \dots + (n-1)$

$$2e \leq n(n-1)$$

$$\text{OR } e \leq \frac{n(n-1)}{2}$$

Thus maximum no. of edges in a simple graph with n

Vertex is $\frac{n(n-1)}{2}$.

3. Suppose G is a non-directed graph with 12 edges. If G has 6 vertices, each of degree 3 & rest have degree less than 3. What is the maximum no. of vertices that G can have?

Ans: We know that $\sum d(v) = 2e$.

Let n be the no. of vertices of G . Then $\sum d(v) \geq$

$$\sum d(v) \geq 6 \times 3 + (n-6) \times 2$$

$$2 \times 12 \geq 18 + (n-6) \times 2$$

$$24 = 18 + 2n - 12$$

$$24 - 6 = 2n$$

$$18 = 2n$$

$$9 = n$$

$$\text{ie } n \geq 9$$

Therefore, maximum no. of vertices in G can have 9.

Applications of graphs:

Graph theory has a wide range of applications in engineering; in physical, social & biological sciences, in linguistics & in other areas.

The following are 10 examples from among 100s of such applications:

- (i) Konigsburg bridge problem.
- (ii) utilises problem

(iii) Seating problem

(iv) Electrical network problem.

(i) Konigsburg bridge problem:

It was a long standing problem until solved by Leonhard Euler in 1756 by means of a graph [Euler is the father of graph theory].

The problem is as follows:

2 islands C & D formed by Pregel river in Konigsburg where connected to each other & to the banks A & B with 7 bridges as shown in the figure (a).

The problem was to start at any of the 4 landmasses of the city A, B, C & D walk over each of 7 bridges exactly once & return to the starting point (without swimming across the river).

Euler represented the situation by means of a graph as shown in figure (b).

The vertices represent the land areas & the edges represent the bridges.

Euler proved that the solution for this problem does not exist.

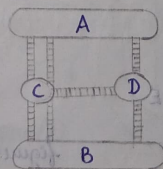


figure (a)

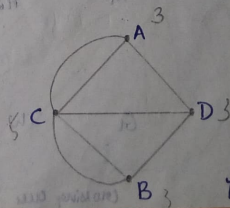


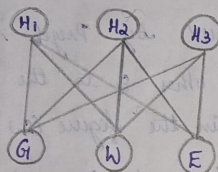
figure (b)

2) Utilities Problem:

The Problem is as follows:

There are 3 houses H_1, H_2, H_3 , each to be connected to each of the 3 utilities: Water (W), Gas (G), & electricity (E) by means of conduits.^{Pipe}

Is it possible to make such connections without any cross over by the conduits.



It violates condition for planarity
 $V - E + F = 2$

3 utilities Problem figure (c)

Now, figure (d) shows how the problem can be represented by a graph.

The conduits are shown as edges while the houses & utilities supply centres are shown as vertices.

The graph cannot be drawn in the plane without edges crossing over. Thus, the answer to this problem is "no".

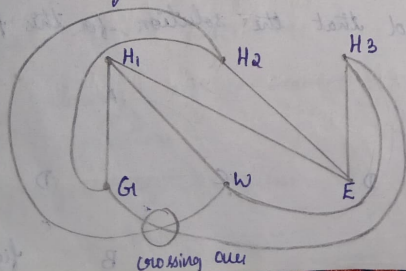


figure (d)

3) Seating problem:

A members of a new club meet at each day for lunch at a round table. They decide to sit such that every member has different neighbours at each lunch.

How many days this arrangements last?

This situation can be represented by a graph with a vertex such that each vertex represents a member & an edge joining a 2 vertices represents the relationship of sitting next to each other.

Figure (e) shows 2 possible seating arrangements. These are:

1-2-3-4-5-6-7-8-9-1 &

1-3-5-2-7-4-9-6-8-1.

There are only 2 more arrangements possible as in figure (f). They are:

1-5-7-3-9-2-8-4-6-1

1-7-9-5-8-3-6-2-4-1

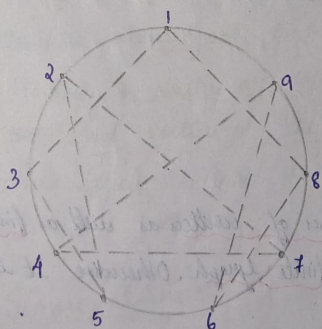


figure (e)

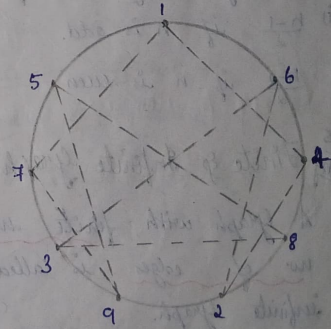


figure (f)

4) Electrical Network Problem:

The Topology of Network is studied by means of its graph. In drawing the graph of an electrical network, the junctions are represented by vertices & branches (which consists of electrical elements) are represented by edges. The electrical network & its graph are shown in figure (g) & figure (h).

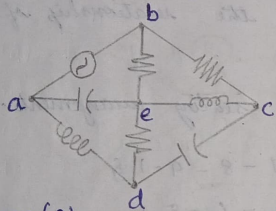


figure (g)

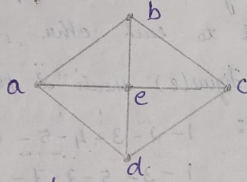


figure (h)

Note:

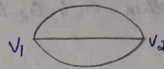
In general, for n people, the no. of such arrangements (seating arrangement problem) is:

- (i) $\frac{n-1}{2}$ if n is odd.
- (ii) $\frac{n-2}{2}$ if n is even.

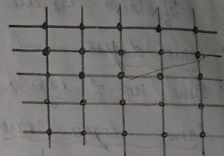
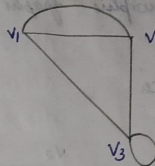
6/1/25 Monday Finite & Infinite Graph:

A graph with finite number of vertices as well as finite no. of edges is called finite graph. Otherwise it is a infinite graph.

Eg:



Finite graph

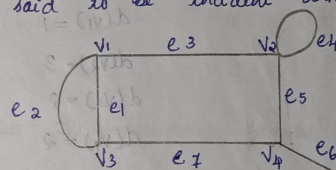


Infinite graph.

7/1/25 Tuesday

Incidence:

When a vertex v_i is an end vertex of some edge e_j , v_i & e_j are said to be incident with (on or to) each other.



Here the edges e_5, e_6, e_7 are incident with the vertex v_4 .

To non-parallel edges are said to be adjacent if they are incident on a common vertex.

In the above graph, e_5 & e_6 are adjacent edges.

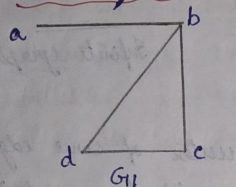
Similarly, 2 vertices are said to be adjacent if they are the end vertices of the same edge. Clearly, from the above figure v_4 & v_5 are adjacent vertices. But v_1 & v_4 are not adjacent.

Isomorphism:

Two graphs G_1 & G_2 are called isomorphic if there is a one-one correspondence between their edges & vertices.

If G_1 & G_2 are isomorphic graphs then G_1 & G_2 have

- same degree
- same no. of edges & vertices
- same degree sequence.



G_1

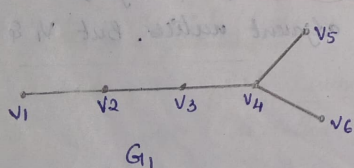
- $d(a) = 1$
- $d(b) = 3$
- $d(c) = 2$
- $d(d) = 2$

→ No. of vertices = 4

→ No. of edges = 5

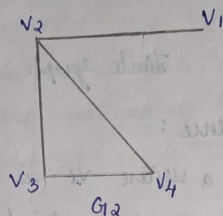
Here G_1 & G_2 are isomorphic since they follow the same degree sequence. [Here in G_1 , b corresponds to vertex v_2 in G_2 & $a \rightarrow v_1$, then $c \rightarrow v_3$, $d \rightarrow v_4$]

Example for non-isomorphic:



G_1

- $d(v_1) = 1$
- $d(v_2) = 1$
- $d(v_3) = 1$
- $d(v_4) = 3$
- $d(v_5) = 1$
- $d(v_6) = 1$

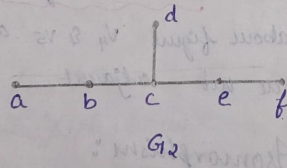


G_2

- $d(v_1) = 1$
- $d(v_2) = 3$
- $d(v_3) = 2$
- $d(v_4) = 2$

No. of vertices = 4

No. of edges = 5



G_2

- $d(a) = 1$
- $d(b) = 1$
- $d(c) = 3$
- $d(e) = 1$
- $d(f) = 1$
- $d(d) = 1$

$$d(v_5) = 1$$

$$d(v_6) = 1$$

No. of edges = 5

No. of vertices = 6

$$d(e) = 2$$

$$d(f) = 1$$

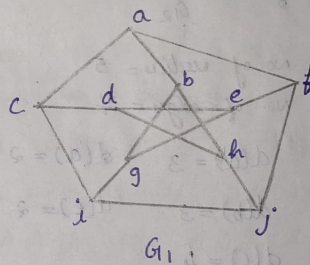
No. of edges = 5

No. of vertices = 6

Here the no. of vertices G_1 & G_2 are same, also the no. of edges of G_1 & G_2 are same & they have the same degree also. But its not isomorphic, since there is no edge corresponding to the edge (e, f) & (b, c) in G_2 . Therefore G_1 & G_2 are not isomorphic.

Check whether the following graphs are isomorphic:

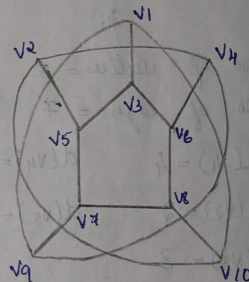
1)



G_1

Ans: No. of vertices = 10
No. of edges = 15

- $d(a) = 3$
- $d(b) = 3$
- $d(c) = 3$



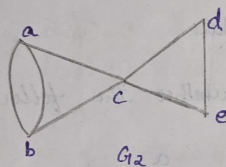
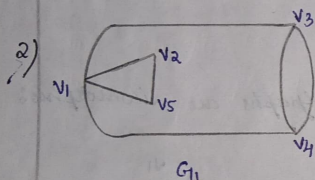
G_2

No. of vertices = 10
No. of edges = 15

- $d(v_1) = 3$
- $d(v_2) = 3$
- $d(v_3) = 3$

$d(d) = 3$	$d(g) = 3$	$d(v_4) = 3$	$d(v_6) = 3$
$d(e) = 3$	$d(h) = 3$	$d(v_5) = 3$	$d(v_7) = 3$
$d(f) = 3$	$d(i) = 3$	$d(v_6) = 3$	$d(v_8) = 3$
	$d(j) = 3$	$d(h) = 3$	

The two graphs G_1 & G_2 have the same no. of vertices & same no. of edges & the same degree & also it follows the same degree sequence. Therefore the graphs G_1 & G_2 are isomorphic.



Ans: No. of vertices = 5
No. of edges = 7

$d(v_1) = 4$	$d(v_4) = 3$
$d(v_2) = 2$	$d(v_5) = 2$
$d(v_3) = 3$	

No. of vertices = 5	$d(a) = 3$	$d(d) = 2$
No. of edges = 7	$d(b) = 3$	$d(e) = 2$
	$d(c) = 4$	

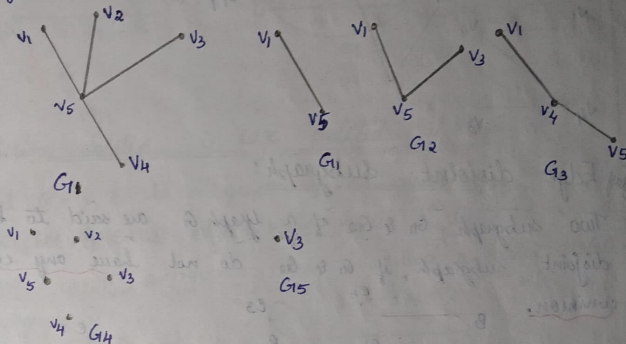
Clearly the graphs G_1 & G_2 are isomorphic since

- The no. of vertices in G_1 & G_2 are same.
- The no. of edges of G_1 & G_2 are same.
- They have the same degree
- They follow the same degree sequence.

Sub-graph:

If $G = (V, E)$ is a graph then $G_1 = (V_1, E_1)$ is a sub-graph of G if $V_1 \neq \emptyset$ & $V_1 \subseteq V$ & $E_1 \subseteq E$ where each edge in E_1 is incident with vertices in V_1 .

Eg:



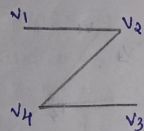
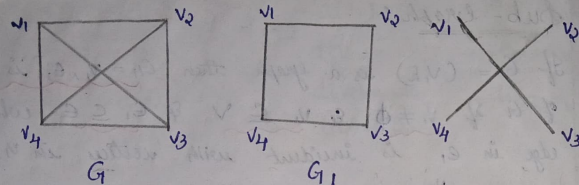
NOTE:

- Every graph is its own subgraph
- A subgraph of a subgraph of G is a subgraph of G .
- A single vertex in a graph G is a subgraph of G .
- A single edge in G together with its end vertices is also a subgraph of G .

Spanning sub-graph:

Let $G = (V, E)$ be a graph & $G_1 = (V_1, E_1)$ is a subgraph of G , then G_1 is called a spanning sub-graph of G , if $V_1 = V$.

Eg:

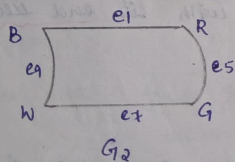
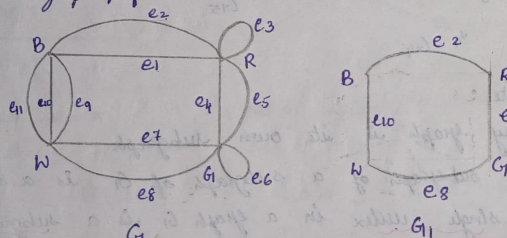


Ellas
Wednesday

Edge disjoint subgraph:

Two subgraph G_1 & G_2 of a graph G are said to be edge disjoint subgraph, if G_1 & G_2 do not have any edge in common.

Eg:

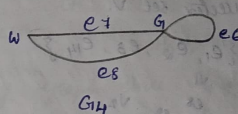
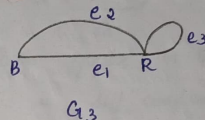


Clearly G_1 & G_2 are subgraphs of G , & since they have no edge in common, we can call G_1 & G_2 are edge disjoint subgraph.

Vertex disjoint subgraph:

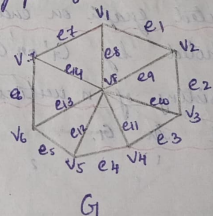
The subgraphs G_1 , that do not have the vertex in common are said to be vertex disjoint subgraph.

Eg:

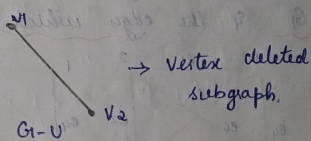


Vertex deleted & edge deleted subgraph:

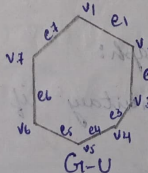
Let $G = V, E$ be a graph. If we delete a subset U of V ($U \subseteq V$) & all edges which have a vertex in U as an end point, or then $G - U$ is called a vertex deleted subgraph of G .



Take $U = \{v3, v4, v5, v6, v7, v8\}$ then we get $G - U$.



Again take $U = \{v8\}$ then we get

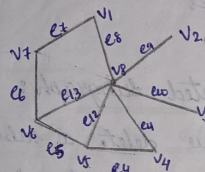


→ Vertex deleted subgraph

Edge deleted subgraph:

If a subset of $E(G)$ is $E - F$ from G then $G - F$ denotes the edge deleted subgraph of G with edge set $E - F$ & vertex set V .

Eg: Take $F = \{e_1, e_2, e_3, e_4\}$

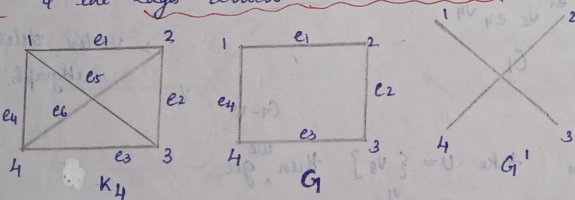


$G - F \rightarrow$ edge deleted subgraph.

Complement of a graph:

Let G be a loop free undirected graph on endvertices then complement of G is denoted by G' or $\bar{G}$ or G^c is the subgraph of K_n , consisting of n vertices in G & the edges which are not in G .

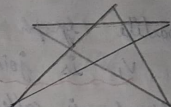
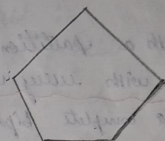
Eg:



Self Complimentary graph:

A Graph G is self complimentary if it is isomorphic to its complement.

Eg:

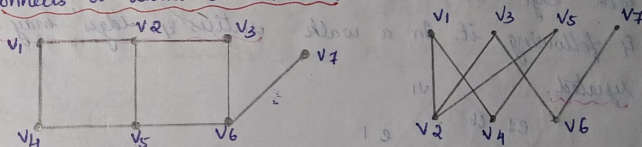


Both graphs G & G^c are isomorphic, the graph G is self complementary graph since G 's no. of vertices, no. of edges & degree of each vertex is same in G & G^c . Also, there is a one-one correspondence between degree sequence.

Bipartite graph:

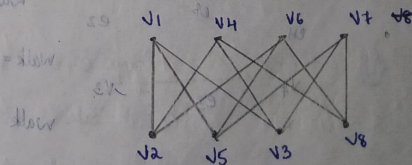
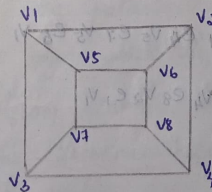
A graph $G(V, E)$ is bipartite, if the vertex set V is partitioned into 2 disjoint subsets V_1 & V_2 i.e. $V_1 \cup V_2 = V$ & $V_1 \cap V_2 = \emptyset$, such that each edge in E , connects a vertex in V_1 & V_2 .

Eg:



$$V_1 = \{V_1, V_3, V_5, V_7\}$$

$$V_2 = \{V_2, V_4, V_6\}$$



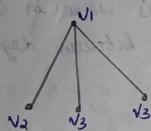
$$V_1 = \{V_1, V_4, V_6, V_7\} \quad V_2 = \{V_2, V_3, V_5, V_8\}$$

Complete bipartite graph:

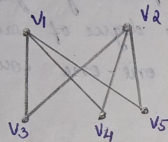
If $G(V, E)$ a bipartite graph with a partition $V = V_1 \cup V_2$ & each vertex in V_1 is joined with every vertex in V_2 . Then the graph is called a complete bipartite graph.

If no. of vertices in $V_1 = m$ & no. of vertices in $V_2 = n$, then complete bipartite graph is denoted by $K_{m,n}$.

Eg:



$K_{1,3}$

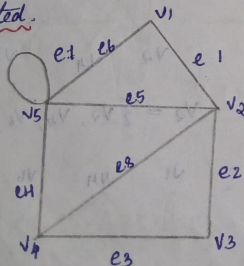


$K_{2,3}$

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Monday

Walk:

A walk is defined as finite alternating sequence of vertices & edges beginning & ending with vertices such that each edge is connected with the vertex preceding & following it. In a walk vertices & edges may be repeated.



$$\text{Walk} = V_1 e_1 V_2 e_2 V_3 e_3 V_4$$

$$\text{Walk} = V_4 e_4 V_5 e_7 V_2 e_5 V_1$$

$$\text{Walk} = V_4 e_8 V_2 e_1 V_1$$

Traill:

→ It is possible for a walk to begin & end at same vertex, such a walk is called a closed walk.

→ A walk that is not closed is called as an open walk, i.e. terminal vertices are distinct.

Eg: $V_1 e_1 V_2 e_5 V_5 e_7 V_5 e_6 V_1$ - closed walk

$V_4 e_8 V_2 e_1 V_1$ - open walk.

Trail:

An alternating sequence of vertices & edges, starting & ending with vertices but no edge repeated, vertex may be repeated.

Eg: $V_2 e_2 V_3 e_3 V_4 e_8 V_2 e_1 V_1$

Path:

An alternating sequence of vertices & edges starting & ending with vertices but no vertices & edges will be repeated.

Eg: $V_2 e_2 V_3 e_3 V_4 e_4 V_5$

→ Length of the path is defined as the no. of edges in the path.

→ An edge which is not a selfloop is of path length one.

→ Selfloop cannot be included in a path.

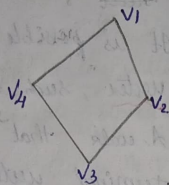
→ The terminal vertices of a path of degree 1 & rest of vertices are of degree 2.

Circuit or cycle:

A closed path is called a circuit.

Eg: $V_1 \in, V_2 \in, V_3 \in, V_4 \in, V_5 \in, V_6 \in$

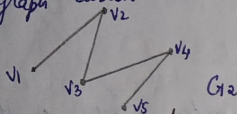
→ Every vertex of degree is 2.



A cyclic graph:

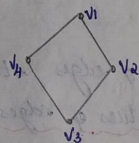
A graph which is not cyclic is called a acyclic graph.

Eg:

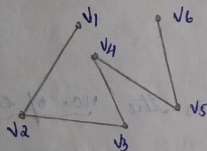


Connected & disconnected graph:

→ A graph is said to be connected if there is a path between any 2 distinct vertices of G . A graph which is not connected is called disconnected graph.



→ disconnected graph.



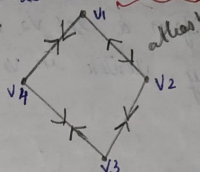
→ connected graph.

Strongly & weakly connected graph:

A directed graph is said to be strongly connected if for every pair of distinct vertices V_i & V_j in G

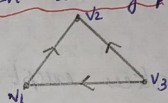
there is a directed path from V_i to V_j & V_j to V_i .

Eg:



A graph which is not strongly connected but its connected then the graph is called weakly connected graph.

Eg:



A disconnected graph consists of 2 or more connected subgraph called components of a graph. The no. of components of a graph is denoted by $k(G)$.

Theorem:

→ A graph G is disconnected if & only if its vertex set V can be partitioned into 2 non-empty disjoint subsets V_1 & V_2 such that there exists no edge in G , whose one end vertex is in subset V_1 & other in subset V_2 .

PROOF:

→ Suppose that such a partition exist, consider 2 arbitrary vertices a & b of G such that $a \in V_1$ & $b \in V_2$. No path can exist b/w the vertices a & b . Hence if a partition exist, G is not connected.

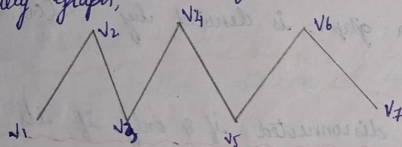
→ Conversely, let G be a disconnected graph, consider a vertex 'a' in G . let V_1 be the set of all vertices that are joined to the vertex a. Since G is

disconnected V_1 does not include all vertices of G .
The remaining vertices will form a set V_2 . No vertex in V_1 is joined by any vertex in V_2 by an edge.

Theorem 2:
→ If a graph (connected & disconnected) has exactly 2 vertices of odd degree, there must be a path joining the 2 vertices.

→ Let G be a graph with all even vertices except the vertex v_1 & v_4 with degree one [odd vertices].

→ For every graph, the no. of odd vertices must be even.



→ Therefore, for every components of a disconnected graph, no graph can have odd no. of odd vertices.

→ Therefore, in graph G , v_1 & v_4 must belong to the same component. Hence must have a path b/w them.

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→ A simple graph with n vertices & k components can have at most $(n-k)(n-k+1)$ edges. *Assume:* let the no. of vertices in each of the k component of a graph be $n_1, n_2, \dots, n_k$.
then, we have $n_1 + n_2 + \dots + n_k = n, n_i \geq 1$.
$$= \sum_{i=1}^k n_i = n$$

Ans: we know that max no. of edges in a simple graph with n vertices is $\frac{n(n-1)}{2}$ which implies $\frac{n_1(n_1-1)}{2} + \frac{n_2(n_2-1)}{2} + \dots + \frac{n_k(n_k-1)}{2} = \sum_{i=1}^k \left[\frac{n_i(n_i-1)}{2} \right]$
$$= \frac{1}{2} \left[\sum_{i=1}^k n_i^2 - \sum_{i=1}^k n_i \right] = \frac{1}{2} \left[\sum_{i=1}^k n_i^2 - n \right] \rightarrow (*)$$

To find the value of $\sum_{i=1}^k n_i^2$ use the following expression
$$\left[\sum_{i=1}^k (n_i - 1) \right]^2 = \left[\sum_{i=1}^k n_i - \sum_{i=1}^k 1 \right]^2 = [n - k]^2$$

$$= n^2 - 2nk + k^2 \rightarrow (**)$$

LHS of $(*)$

$$\left[\sum_{i=1}^k (n_i - 1) \right]^2 = [(n_1 - 1) + (n_2 - 1) + \dots + (n_k - 1)]^2$$
$$= (n_1 - 1)^2 + (n_2 - 1)^2 + \dots + (n_k - 1)^2 + 2(n_1 - 1)(n_2 - 1) + 2(n_2 - 1)(n_3 - 1) + \dots + 2(n_{k-1} - 1)(n_k - 1)$$

ie $\sum_{i=1}^k (n_i - 1)^2 + \text{non negative cross terms}$.

Now equating LHS & RHS of eq $(*)$. we get,

$$\left[\sum_{i=1}^k (n_i - 1)^2 \right] + \text{non negative cross terms} = n^2 - 2nk + k^2$$

Then $\left[\sum_{i=1}^k (n_i - 1)^2 \right] \leq n^2 - 2nk + k^2$

ie $\sum_{i=1}^k n_i^2 - 2 \sum_{i=1}^k n_i + \sum_{i=1}^k 1 = n^2 - 2nk + k^2$

ie $\sum_{i=1}^k n_i^2 - 2 \sum_{i=1}^k n_i + \sum_{i=1}^k 1 = n^2 - 2nk + k^2$

$$\sum_{i=1}^k n_i^2 = n^2 - 2nk + k^2 + 2n - k$$

$$\sum_{i=1}^k n_i^2 - 2n + k = n^2 - 2nk + k^2$$

$$\sum_{i=1}^k n_i^2 \leq n^2 - 2nk + k^2 + 2n - k$$

Now substitute in eqⁿ we get the no. of edges is,

$$= \leq \frac{1}{2} (n^2 - 2nk + k^2 + 2n - k - n)$$

$$= \frac{1}{2} [n^2 - nk - nk + k^2 + 2n - k - n]$$

$$= \frac{1}{2} [n(n-k+1) - k(n-k+1)]$$

$$= \frac{1}{2} [(n-k)(n-k+1)] \text{ i.e. } \frac{(n-k)(n-k+1)}{2}$$

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Monday

Module-2

Eulerian & Hamiltonian Graphs

Euler graph:

A graph which contains an euler line is called an euler graph.

Euler line:

A closed walk which should pass through all the edges of G exactly once.

Euler Path:

It is a path that passes through all the edges of the graph almost once i.e. starting & ending vertices are different & no vertex appears more than once.

Euler circuit:

It is a circuit which passes through all the edges of the graph almost once. Here no vertex will appear more than once except starting & ending vertices.

Points to remember for euler line:

- closed walk
- Passes through all the edges of the graph exactly once.
- vertices can be repeated.